

Blueprint Completo de la Base de Datos — AMS / MagEAM

Versión: 1.0 · 2026-05-12

BD: SQLite (single-file `data/ocp_maintenance.db`)

ORM: SQLAlchemy 2.0 + Pydantic

Modelos: 67 tablas en `api/database/models.py`

Sincronización: auto-create_all en boot (sin Alembic todavía)

1. Vista general — 67 tablas en 10 dominios

Dominio	Tablas	#
Core (scoping multi-tenant)	users, plants, hierarchy_nodes, criticality_assessments, audit_log	5
RCM (análisis de fallas + estrategias)	functions, functional_failures, failure_modes, maintenance_tasks, work_packages, fmeca_worksheets, rca_analyses, capa_items	8
Capture → WR → OT (flujo principal)	field_captures, work_requests, managed_work_orders, work_orders, backlog_items, optimized_backlog, planner_recommendations, sync_conflicts	8
Resources (personal + materiales + equipos)	workforce, support_equipment, inventory_items, contractors, contractor_crews, equipment_3d_models	6
Scheduling (programa semanal + shutdowns)	weekly_programs, shutdown_calendar, shutdown_events, preparativos_ot	4
Execution (cierre + reviews)	execution_checklists, work_assignments, equipment_handovers, post_maintenance_reviews, time_logs	5
KPIs + Analytics	health_scores, kpi_metrics, failure_predictions, variance_alerts, planning_kpi_snapshots, de_kpi_snapshots, reports	7
Confiabilidad avanzada	moc_requests, spare_part_analyses, rbi_assessments, improvement_actions	4
IA + Agentes	ai_sessions, ai_interactions, agentic_executions, troubleshooting_diagnostics, troubleshooting_sessions, expert_cards, expert_consultations, expert_contributions	8
Integración + DMS + SAP	sap_upload_packages, sap_sync_log, dms_documents, import_history	4
Feedback + entregables + or-system	user_feedback, detailed_feedback, ai_feedback, dedup_negative_pairs, deliverables, notifications, or_projects, or_deliverables	8

Total: 67 tablas

2. Tablas Core con relaciones lógicas a las operacionales

2.1 plants — origen multi-tenant

Campo	Tipo	Notas
`plant_id`	string PK	Ej. GOLDFIELDS-SN, OCP-JFC1
`name`	string	Nombre legible (anonimizado a "Planta Minera Cliente")
`location`	string	Geográfica
`language`	string	Idioma default UI

`plant_id` se propaga (sin FK formal) a: `users` , `workforce` , `hierarchy_nodes` , `work_requests` , `managed_work_orders` , `field_captures` , `audit_log` , etc.

2.2 users

Campo	Tipo	Notas
`user_id`	string PK (uuid)	
`username`	string unique	
`password_hash`	string	bcrypt
`role`	string	admin / manager / planner / engineer / supervisor / tecnico
`plant_id`	string	Scope multi-tenant
`token_version`	int	bump invalida todos los JWT del usuario
`is_active`	bool	

2.3 hierarchy_nodes — estructura del activo

Campo	Tipo	Notas
`node_id`	string PK	
`parent_node_id`	string (self-ref)	Plant > Area > System > Equipment
`node_type`	string	PLANT / AREA / SYSTEM / EQUIPMENT
`tag`	string	TAG SAP
`code`	string	Código simplificado
`name`	string	
`level`	int	1-4
`plant_id`	string	
`criticality`	string	Snapshot último assessment

2.4 criticality_assessments

Campo	Tipo	Notas
`assessment_id`	PK	
`node_id`	FK hierarchy_nodes	
`risk_class`	enum	I_LOW / II_MEDIUM / III_HIGH / IV_CRITICAL
`band`	string	B / A / A+ / AA
`status`	string	DRAFT / APPROVED
`ai_generated`	bool	
`ai_confidence`	float	0-1

2.5 audit_log — universal append-only

Campo	Tipo	Notas
`id`	int PK autoincrement	
`timestamp`	DateTime indexed	UTC
`entity_type`	string indexed	work_request / managed_work_order / user / settings / ... (14 entidades cubiertas)
`entity_id`	string	Generic foreign reference
`action`	string	CREATE / UPDATE / TRANSITION / APPROVE / DELETE / LOGIN / ...
`payload`	JSON	Datos completos del cambio
`user`	string	`` o `agent:`
`ip_address`	string	Para LOGIN
`user_agent`	string	Para LOGIN

Política SF-660: append-only, 2 años activo, 5 años archivo, eliminación con notificación.

3. Dominio RCM — Análisis de fallas

3.1 functions

Campo	Tipo
`function_id` PK	string
`node_id` FK hierarchy_nodes	string
`function_type`	string (PRIMARY / SECONDARY / PROTECTION)
`description`	text

3.2 functional_failures

Campo	Tipo
`failure_id` PK	string
`function_id` FK functions	string

`failure_type`	string (TOTAL / PARTIAL / INTERMITTENT)
`description`	text

3.3 failure_modes

Campo	Tipo
`failure_mode_id` PK	string
`functional_failure_id` FK NOT NULL	string
`mechanism`	string
`cause`	string
`failure_pattern`	string A-F
`failure_consequence`	string
`strategy_type`	string (CONDITION_BASED / SCHEDULED_DISCARD / RUN_TO_FAILURE / REDESIGN)
`ai_generated`	bool
`ai_confidence`	float
`existing_task_source`	string nullable

3.4 maintenance_tasks

Campo	Tipo
`task_id` PK	string
`failure_mode_id` FK	string
`task_type`	string (INSPECT / TEST / REPLACE / OVERHAUL / MONITOR)
`interval`	string
`resources`	JSON

3.5 work_packages

Campo	Tipo
`package_id` PK	string
`tasks`	JSON list of task_ids
`specialty`	string
`duration_hours`	float
`frequency`	string

Cada work_package debe contener 7 elementos mandatorios: permit, LOTO, materials, checklist, JRA, procedure, WO.

3.6 fmeca_worksheets

Campo	Tipo
`worksheet_id` PK	string
`node_id`	string

`status`	string (DRAFT / APPROVED)
`generated_by`	string
`approval_chain`	JSON

3.7 rca_analyses

Campo	Tipo
`rca_id` PK	string
`wo_id` nullable	string
`method`	string (5why / fishbone / fault_tree)
`root_causes`	JSON
`recommendations`	JSON
`approval_status`	string

3.8 capa_items (PDCA)

Campo	Tipo
`capa_id` PK	string
`source_type`	string (RCA / AUDIT / FMECA / USER)
`pdca_phase`	string (PLAN / DO / CHECK / ACT)
`status`	string
`owner_id`	string
`due_date`	datetime

4. Flujo principal: Capture → WR → OT

4.1 field_captures

Campo	Tipo	Notas
`capture_id`	string PK	
`technician_id`	string	
`capture_type`	string	VOICE / TEXT / IMAGE / VOICE+IMAGE
`language`	string	en / es / fr / ar
`raw_text`	text nullable	
`raw_voice_text`	text nullable	Transcripción audio
`images`	JSON list	URLs `/api/v1/capture/photos/...`
`location_hint`	string	
`gps_lat` / `gps_lng`	float	

4.2 work_requests (AV-NNNNN)

Campo	Tipo	Notas
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`request_id`	string PK	
`source_capture_id`	FK field_captures	nullable
`status`	string	DRAFT → APROBADO / RECHAZADO → LINKED
`equipment_id`	string	
`equipment_tag`	string	
`equipment_confidence`	float	AI match score
`ai_classification`	JSON	priority, equipment, suggestions
`image_analysis`	JSON	anomalies, severity, component
`priority_code`	string	P1 / P2 / P3 / P4
`work_class`	string	PROGRAMADO / NO_PROGRAMADO
`sla_deadline`	datetime	
`created_by`	string	
`approver_id`	string	Quien aprobó
`approved_at`	datetime	
`approval_comment`	text	
`rejection_reason`	text	
`notification_type`	string	A1 (SAP)
`reported_by`	string	Autor del aviso SAP
`documents`	JSON	adjuntos
`support_equipment`	JSON	
`circumstances`	text	SAP campo 8
`aviso_coding`	string	M001-M005, P001-P003
`planning_group`	string	M01-M05, P01-P03
`area_empresa`	string	SEC, HUM, RIP, etc.
`version`	int	Optimistic lock

4.3 managed_work_orders (OT-2026-NNNNN) — TABLA CENTRAL

Tabla más grande. **55 campos + 8 JSON arrays embebidos.**

Campos básicos

Campo	Tipo	Notas
`wo_id`	string PK	uuid
`wo_number`	string unique	OT-2026-NNNNN
`work_request_id`	string nullable	sin FK formal (fast-track P1/P2 puede ser standalone)
`plant_id`	string	
`equipment_id`	string	
`equipment_tag`	string	
`technical_location`	string nullable	TL SAP

Planning

Campo	Tipo	Notas
`wo_title`	string nullable	SF-507 título arrastrado del WR
`description`	text	
`description_history`	JSON list	SF-653 append-only
`wo_type`	string	PM01 / PM02 / PM03 (correctivo/preventivo/mejora)
`priority_code`	string	P1 / P2 / P3 / P4
`work_class`	string	PROGRAMADO / NO_PROGRAMADO
`is_fast_track`	bool	P1/P2 bypass planning
`planning_group`	string nullable	
`work_center`	string nullable	

JSON arrays embebidos (no son tablas separadas)

Campo	Estructura interna
`operations`	`[[{seq, description, specialty, hours, quantity, completion_pct, actual_hours, task_type}]]`
`materials`	`[{code, description, quantity, unit, reservation_code, status, cost_unit}]`
`tools`	`[{tool_name, qty}]`
`support_equipment`	`[{tag, name, equipment_type, hours, notes, scope}]`
`documents`	`[{name, url, type}]`
`assigned_workers`	`[{worker_id, name, specialty}]`
`post_closure_review`	`{outcome, lessons_learned[], recommendations[]}`
`labour_summary`	`{total_hours, specialties: [{name, hours}]}`

Scheduling

Campo	Tipo
`planned_start` / `planned_end`	datetime
`actual_start` / `actual_end`	datetime
`estimated_hours`	float
`actual_hours`	float
`shift`	string (DAY / NIGHT)

Workflow

Campo	Tipo	Valores
`status`	string	CREADO → LIBERADO → PLANIFICADO → EN_PROGRAMACION → PROGRAMADO → EN_EJECUCION → CERRADO (también REPROGRAMADO, CANCELADO)
`planned_by`	string nullable	
`released_by`	string nullable	

`released_at`	datetime nullable	
`closed_by`	string nullable	
`closed_at`	datetime nullable	

Cierre con firma (SF-570 + pre-close gates)

Campo	Tipo	Notas
`closed_by_signature`	string nullable	Nombre tipeado
`closed_by_pin_hash`	string(16)	sha256 truncated 16 chars del PIN
`closure_notes`	string(500)	
`closure_audio_url`	string(500)	SF-500
`contractor_crew_id`	string nullable	

Costos

Campo	Tipo
`budget_approved` / `budget_amount`	bool / float
`labor_cost` / `material_cost` / `external_cost`	float
`planned_material_cost` / `planned_external_cost`	float
`actual_total_cost`	float

Otros

Campo	Tipo	Notas
`completion_pct`	float	
`reservation_code`	string nullable	SAP BOM reserva
`cancellation_reason` / `cancellation_type`	string nullable	
`absorbed_by_wo_id`	string nullable	self-ref: OT PM03 absorbe PM01/PM02
`version`	int	Optimistic lock SF-562
`created_at` / `updated_at`	datetime	

4.4 backlog_items

Campo	Tipo
`backlog_id` PK	string
`work_request_id`	string
`wo_id` nullable	string
`equipment_id` / `equipment_tag`	string
`priority`	string
`wo_type`	string
`status`	string (READY / BLOCKED / OPTIMIZED / SCHEDULED)
`blocking_reason`	string nullable
`estimated_hours`	float
`specialties`	JSON list

`materials_ready`	bool
`shutdown_required`	bool
`age_days`	int
`plant`	string
`group_id`	string nullable

4.5 preparativos_ot (SF-662 — NUEVA SP7)

Campo	Tipo	
`prep_id` PK	string	
`wo_id` indexed	string	
`plant_id` indexed nullable	string	
`item_code`	string	
`item_desc`	text nullable	
`qty`	float	
`unit`	string nullable	
`status` indexed	PENDIENTE / DESPACHADO / EN_TRANSITO / RECIBIDO / ANOMALIA	
`dispatched_at` / `dispatched_by`	datetime / string	
`received_at` / `received_by`	datetime / string	
`conforme`	bool nullable	firma de aceptación al recibir
`layout_url`	string(500)	PDF disposición patio
`notes`	text	append-only con timestamp

5. Resources

5.1 workforce

Campo	Tipo	Notas
`worker_id` PK	string	
`name`	string	
`specialty`	string	MECANICO / ELECTRICO / INSTRUMENTISTA / SOLDADURA / etc.
`shift`	string	MORNING / AFTERNOON / NIGHT
`plant_id`	string	
`available`	bool	
`certifications`	JSON	
`competency_level`	string	A / B / C / D
`years_experience`	int	

`equipment_expertise`	JSON	
`skills`	JSON	
`shift_pattern`	string	5x2 / 4x3 / 7x7 / 14x14 / continuous / abc_8h
`shift_cycle_start`	string	ISO date
`absence_reason` / `absence_until`	string nullable	

5.2 support_equipment

Campo	Tipo	Notas
`equipment_id` PK	string	
`tag`	string unique	
`name`	string	
`equipment_type`	string	MOBILE_CRANE / SCAFFOLDING / BRIDGE_CRANE / etc.
`plant_id`	string	
`capacity`	float	
`certification`	string	
`available`	bool	
`alias`	JSON	Multilingual (Grúa móvil ↔ Mobile crane)

5.3 inventory_items (catalog materiales)

Campo	Tipo
`material_code` PK	string
`warehouse_id`	string
`description`	string
`quantity_on_hand`	int
`quantity_reserved`	int
`quantity_available`	int
`min_stock` / `reorder_point`	int
`last_movement_date`	date
`vendor`	string

5.4 contractors + contractor_crews

`contractors`	`contractor_crews`
contractor_id PK	crew_id PK
name	contractor_id FK
specialty	crew_name
contact / phone	members (JSON)
status	active

6. Scheduling

6.1 weekly_programs

Campo	Tipo
`program_id` PK	string
`plant_id`	string
`week_number / year`	int / int
`status`	DRAFT / FINALIZED / ACTIVE / COMPLETED
`wo_ids`	JSON list
`total_planned_hh`	float
`specialty_load`	JSON

6.2 shutdown_calendar + shutdown_events

`shutdown_calendar`	`shutdown_events`
shutdown_id PK	event_id PK
plant_id	shutdown_id FK
start_date / end_date	wo_id
shutdown_type	duration
areas (JSON)	critical_path
description	

7. Execution + Closure

7.1 execution_checklists

Campo	Tipo
`checklist_id` PK	string
`wo_id`	string
`operation_seq`	int
`items`	JSON list
`status`	string
`signed_by`	string

7.2 work_assignments

Campo	Tipo
`assignment_id` PK	string
`wo_id`	string
`worker_id`	string
`specialty`	string

`shift`	string
`assigned_hours`	float
`confirmed`	bool

7.3 equipment_handovers

Campo	Tipo
`handover_id` PK	string
`wo_id`	string
`from_shift` / `to_shift`	string
`notes`	text
`signature`	string
`photos`	JSON list

7.4 post_maintenance_reviews

Campo	Tipo
`review_id` PK	string
`wo_id`	string
`outcome`	string
`effectiveness`	string
`lessons_learned`	text
`recommendations`	text
`approved_by`	string

7.5 time_logs

Campo	Tipo
`log_id` PK	string
`wo_id`	string
`worker_id`	string
`operation_seq`	int
`hours`	float
`logged_at`	datetime

8. KPIs

8.1 kpi_metrics

Campo	Tipo
`metric_id` PK	string
`plant_id`	string
`period`	string (week / month / quarter)

`category`	string
`values`	JSON (MTBF, MTTR, MTBM, availability, etc.)

8.2 planning_kpi_snapshots

Campo	Tipo
`snapshot_id` PK	string
`plant_id`	string
`schedule_compliance`	float
`schedule_adherence`	float
`backlog_age` / `backlog_size`	int
`reactive_ratio`	float
`wrench_time`	float

8.3 de_kpi_snapshots (Defect Elimination)

Campo	Tipo
`snapshot_id` PK	string
`plant_id`	string
`period_start` / `period_end`	date
`events_reported` / `events_required`	int / int
`meetings_held` / `meetings_required`	int / int
`actions_implemented` / `actions_planned`	int / int
`savings_achieved` / `savings_target`	float / float
`failures_current` / `failures_previous`	int / int
`maturity_level`	INITIAL / DEVELOPING / ESTABLISHED / OPTIMIZING
`program_score`	float 0-100

8.4 health_scores

Campo	Tipo
`score_id` PK	string
`equipment_id`	string
`score`	float 0-100
`trend`	string (improving / stable / degrading)
`recommendations`	text

8.5 failure_predictions

Campo	Tipo
`prediction_id` PK	string
`equipment_id`	string
`predicted_date`	datetime
`confidence`	float

`failure_mode_id`	string
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8.6 variance_alerts

Campo	Tipo
`alert_id` PK	string
`wo_id`	string
`alert_type`	HH_VARIANCE / COST_VARIANCE / SCHEDULE
`severity`	string
`resolved`	bool

9. IA + Agentes

9.1 ai_sessions

Campo	Tipo
`session_id` PK	string
`user_id`	string
`agent_type`	string
`status`	string
`token_count`	int
`cost_usd`	float

9.2 ai_interactions

Campo	Tipo
`interaction_id` PK	string
`session_id` FK	string
`prompt`	text
`response`	text
`model`	string (claude-opus-4-6 / claude-sonnet-4-5 / etc.)
`tokens_in` / `tokens_out`	int / int
`latency_ms`	int
`feedback`	int (1 = thumbs up, -1 = down)

9.3 agentic_executions

Campo	Tipo
`exec_id` PK	string
`agent_id`	string
`skill_used`	string
`input` / `output`	JSON
`tokens_used`	int

`duration_ms`	int
---------------	-----

9.4 troubleshooting_sessions + troubleshooting_diagnostics

Pares para sesiones de troubleshooting (P1/P2) con árbol de decisión + match a casos previos.

9.5 expert_cards + expert_consultations + expert_contributions

Sistema de knowledge base con expertos: tarjetas (casos resueltos), consultas (preguntas abiertas), contribuciones (respuestas).

10. Integración + DMS + SAP

10.1 dms_documents

Campo	Tipo	Notas
`doc_id` PK	string	
`document_number` unique indexed	string	DOC-NNNNNNN
`document_type` indexed	string	DWG / MAF / MAN / PRO / CHK
`document_desc`	text	
`version`	int	
`sap_func_loc / sap_func_loc_short` indexed	string nullable	
`equipment_name / eqart`	string nullable	
`file_path`	string(300) nullable	
`created_date / created_by`	string	
`status`	string	Activo / Inactivo

10.2 sap_upload_packages

Campo	Tipo
`package_id` PK	string
`wo_ids`	JSON list
`status`	DRAFT / READY / UPLOADED / FAILED
`format`	CSV / JSON
`generated_by`	string
`created_at`	datetime

10.3 sap_sync_log

Campo	Tipo
`sync_id` PK	string
`entity_type / entity_id`	string / string
`direction`	PUSH / PULL

`status`	PENDING / SUCCESS / FAILED
`payload`	JSON
`error_message`	text
`retry_count`	int

10.4 import_history

Campo	Tipo
`import_id` PK	string
`source_file`	string
`entity_type`	string
`row_count`	int
`errors`	JSON
`imported_by`	string

11. Convenciones del schema

Convención	Detalle
PK = `_id`	wo_id, worker_id, capture_id, etc. (UUID string)
Timestamps	created_at + updated_at auto via SQLAlchemy onupdate
Soft-delete	NO existe — usar status=CANCELLED
Multi-tenant	`plant_id: str` en todas las tablas operacionales
JSON embebido	Para arrays variables (operations, materials, etc.)
Optimistic lock	Campo `version: int` en managed_work_orders y WR
Audit trail	audit_log tabla universal con entity_type + entity_id
FK explícitas	Solo donde el borrado cascadeante es deseable
FK lógicas	Para resiliencia: plant_id, worker_id, material_code (string sin constraint)

12. Índices clave

```
```sql
```

```
CREATE INDEX idx_audit_log_entity_type ON audit_log(entity_type);
```

```
CREATE INDEX idx_audit_log_entity_id ON audit_log(entity_id);
```

```
CREATE INDEX idx_audit_log_timestamp ON audit_log(timestamp);
```

```
CREATE INDEX idx_wo_status ON managed_work_orders(status);
```

```
CREATE INDEX idx_wo_plant_id ON managed_work_orders(plant_id);
```

```
CREATE INDEX idx_wo_wo_number ON managed_work_orders(wo_number);
```

```
CREATE INDEX idx_wo_priority ON managed_work_orders(priority_code);
```



```

CREATE INDEX idx_wr_status ON work_requests(status);
CREATE INDEX idx_wr_equipment ON work_requests(equipment_id);

CREATE INDEX idx_hierarchy_node_type ON hierarchy_nodes(node_type);
CREATE INDEX idx_hierarchy_tag ON hierarchy_nodes(tag);
CREATE INDEX idx_hierarchy_plant_id ON hierarchy_nodes(plant_id);

CREATE INDEX idx_preparativos_wo_id ON preparativos_ot(wo_id);
CREATE INDEX idx_preparativos_status ON preparativos_ot(status);
...

```

## 13. State machines

### 13.1 WorkRequest status

Estado	Siguiente	Transición
DRAFT	APROBADO	supervisor valida con action=APPROVE
DRAFT	RECHAZADO	supervisor valida con action=REJECT (terminal)
APROBADO	LINKED	cuando se crea OT from-wr

### 13.2 ManagedWorkOrder status (SF-578 cadena)

Estado	Siguientes válidos	Endpoint dedicado
CREADO	LIBERADO, PROGRAMADO (fast-track P1/P2)	`/draft`, `/release`
LIBERADO	PLANIFICADO	`/plan`
PLANIFICADO	EN_PROGRAMACION	PUT generic
EN_PROGRAMACION	PROGRAMADO	`/schedule`
PROGRAMADO	EN_EJECUCION, REPROGRAMADO	`/start`, `/reschedule`
REPROGRAMADO	PROGRAMADO	`/schedule`
EN_EJECUCION	CERRADO, REPROGRAMADO	`/close`, `/reschedule`
CERRADO	(terminal)	—
CANCELADO	(terminal)	`/cancel`

### 13.3 Preparativos OT status (SF-662)

Estado	Siguientes	Requisitos
PENDIENTE	DESPACHADO	—
DESPACHADO	EN_TRANSITO, ANOMALIA	—
EN_TRANSITO	RECIBIDO, ANOMALIA	conforme obligatorio al recibir

RECIBIDO	(terminal)	—
ANOMALIA	DESPACHADO	reintento

## 14. Compatibilidad SAP PM

### 14.1 work\_requests → Aviso IH01

Campo AMS	Campo SAP	Significado
`aviso_coding`	QMART (10)	Tipo aviso (M001-M005, P001-P003)
`notification_type`	QMNUM tipo	A1=Aviso Mantto
`planning_group`	INGRP (10)	Grupo planificación
`work_center`	ARBPL (20)	Puesto de trabajo
`technical_location`	TPLNR (100)	Ubicación técnica
`area_empresa`	ABCKZ (10)	Indicador ABC empresa
`reported_by`	QMNAM	Autor del aviso
`circumstances`	LTXT	Texto largo descripción

### 14.2 managed\_work\_orders → Orden IW21

Campo AMS	Campo SAP	Significado
`wo_number`	AUFNR	Número de orden
`wo_type`	AUART	Clase de orden (PM01/02/03)
`priority_code`	PRIOK	Prioridad
`equipment_id`	EQUNR	Número de equipo
`planned_start / planned_end`	GSTRP / GLTRP	Fechas plan
`actual_start / actual_end`	GSTRI / GLTRI	Fechas real

Mapping completo en `sap_mock/data/avisos_campos_blueprint.xlsx` .

## 15. Operaciones administrativas

### Auto-creación (sin Alembic todavía)

```
```python
```

api/main.py

```
from api.database.connection import Base, engine

Base.metadata.create_all(engine) # crea tablas faltantes al boot
```
```

Backup recomendado

```
```bash
```

SQLite backup atómico (lockless)

```
sqlite3 data/ocp_maintenance.db ".backup data/backup_$(date +%Y-%m-%d).db"
```
```

Limpiar audit > 5 años

```
```bash
python scripts/purge_audit_log_5y.py --dry-run # ver qué borraría
python scripts/purge_audit_log_5y.py --confirm --export-csv X.csv
```
```

Re-indexar RAG (LanceDB separado de SQLite)

```
```bash
python scripts/reindex_rag.py
```
```

16. Tablas que faltan (SP8+)

- `cost_centers` (OE tanda) — opex/capex por CeCo
- `expense_classes` (OE) — clases de gasto SAP
- `failure_symptom_catalog` (SAP-style) — síntomas codificados
- `failure_cause_catalog` (SAP-style) — causas codificadas
- `wrench_time_observations` — KPI tradicional
- `training_records` — horas capacitación
- `safety_incidents` — integración HSE
- `equipment_telemetry` — futuro SCADA real

17. Diferencias `work_orders` (legacy) vs `managed_work_orders`

| Aspecto   | `work_orders` (Fase 1)         | `managed_work_orders` (actual)       |
|-----------|--------------------------------|--------------------------------------|
| Uso       | Legacy, sin lifecycle complejo | OT principal SAP-PM completo         |
| Campos    | Mínimos (8)                    | 55+ campos + JSON embedded           |
| Lifecycle | Solo created / closed          | 7 estados con transiciones validadas |
| Auditoría | No                             | Full audit_log integration           |

|                          |                            |                          |
|--------------------------|----------------------------|--------------------------|
| Fast-track               | No                         | P1/P2 bypass planning    |
| Closure gates            | No                         | 6 pre-close gates SF-570 |
| <b>**Recomendación**</b> | <b>**Deprecar**</b> en SP9 | Único modelo a usar      |

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**Total: 67 tablas · 562 endpoints · SQLite single-file ~50MB en producción**

**Commit:** `cd83670`